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Rolling bearing fault diagnosis method based on DSICNN under strong noise background

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ABSTRACT

To address the issues of inadequate fault diagnosis accuracy and suboptimal generalisation performance of rolling bearings in the presence of significant noise and varying operational conditions, a fault diagnosis approach utilising a dual-stream interactive convolutional neural network (DSICNN) is presented. To fully leverage the fault characteristics in vibration signals, both time-domain and frequency-domain signals are concurrently employed as inputs to the neural network. Then, dual attention mechanisms are presented, among which the Dynamic Weighted Channel Attention Mechanism (DWCAM) dynamically calibrates the channel weights of different inputs based on the importance of different channels, and the Aggregated Spatial Self Attention Mechanism (ASSAM) assigns greater weights to important regions while enhancing feature expression ability. Meanwhile, an arctangent linear unit function (AT-LU) is constructed to improve the problem of information loss contained in the signal when the linear rectifier unit has negative input. Finally, the signals are input into DSICNN, and the features are extracted more fully by interacting with two information streams. The model is trained and evaluated against other fault diagnosis models. The experimental results indicate that the suggested method exhibits superior classification performance, generalisation capability and robustness in the presence of significant noise and varying operating conditions.

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Fault diagnosis; fast Fourier transform; dual-stream; attention mechanism; rolling bearings

1. Introduction

Rotating machinery has been widely used in modern industry, and its operating status directly affects the entire machine, including reliability and stability. Therefore, the fault diagnosis of rotating machinery has attracted the attention of many scholars [1]. Rolling bearings are essential components of rotating machinery, significantly impacting aerospace [2], CNC machine tools [3], wind turbines [4] and various other sectors and are known as the 'joints of industry [5]'. Rolling bearings usually operate under extreme conditions such as variable speed, variable load and high temperature, and are one of the most vulnerable mechanical components. About 45%–55% of rotating mechanical failures result from damage to rolling bearings [6]. Therefore, the detection of faults in rolling bearings is crucial for ensuring the normal operation of rotating machinery and

the efficient and safe production of modern industry. In practical engineering, the fault characteristics of vibration signals are prone to be masked by environmental noise, making it difficult to extract and distinguish bearing fault types. Therefore, achieving accurate and stable fault diagnosis in noisy environments is a very challenging task.

Vibration signals, being a type of periodic time series data [7], are frequently utilised for the defect detection of mechanical equipment. Traditional fault diagnosis methods rely on empirical formulas and mathematical modelling, with signal processing as the core, and use physics and engineering laws to construct system models [8]. However, this method typically requires a significant amount of expert knowledge and has poor generality. The fault diagnosis methods based on traditional machine learning use signal processing techniques to extract key features and use them as inputs to train traditional machine learning algorithms for fault classification. Xiang et al. [1] rendered a personalised fault diagnosis method using the finite element method (FEM), wavelet packet transform (WPT) and support vector machine (SVM) to detect faults. Compared to traditional methods, it reduced reliance on physical models and could handle features from more dimensions. However, feature extraction relies on manual design, and insufficient or poorly created features could not bring satisfactory diagnostic results.

In recent years, deep learning-based diagnostic algorithms have gained widespread use in fault diagnosis due to their ability to automatically extract high-order features from data, reducing the reliance on manually designed features and enhancing model robustness and generalisation [9], such as Convolutional Neural Network (CNN), Generative Adversarial Network (GAN) and Deep Belief Network (DBN). Some scholars have achieved good diagnostic results using these methods. Gao et al. [10] presented numerical simulation and Generative Adversarial Network (GAN) to expand the fault sample for a new method of fault detection. Lou et al. [11] proposed a domain adaptive fault diagnosis method based on personalised enhancement of diagnostic method to address the issue of insufficient and incomplete training samples.

Among deep learning methods, CNNs have gained particular attention for their ability to process periodic signals and learn features from mechanical vibration signals. These networks are widely used in rolling bearing fault diagnosis [12]. Shi et al. [13] rendered an enhanced lightweight multi-scale convolutional neural network (CNN) for fault diagnosis of rolling bearings, which had good fault feature extraction and noise resistance capabilities. Liang et al. [14] put forward a novel approach utilising a one-dimensional dilated convolutional network with residual connections for diagnosing rolling bearing problems across varying noise conditions and loads. Chen et al. [15] constructed a multi-scale CNN-LSTM neural network model and a deep residual learning model, which effectively extracted fault features hidden in noisy signals. Xu et al. [16] proposed an improved multi-scale branch convolutional neural network for rolling bearing fault diagnosis, improving the accuracy of bearing diagnosis effectively. The aforementioned models utilise raw vibration signals as input data and have demonstrated efficacy in diagnosing rolling bearing defects in noisy situations. However, some scholars convert the original time-domain signals into frequency-domain signals as inputs of neural networks in order to enhance the noise resistance and robustness of the model. Hakim et al. [17] and Yang et al. [18] proposed one-dimensional convolutional neural networks (1D-CNN) utilising frequency domain signal processing and dynamic convolutional capsule networks (DC-CapNets) for rolling bearing diagnostics, respectively. Both

methods took the original vibration signal after Fast Fourier Transform (FFT) as the model input and demonstrated effective defect diagnostic performance in noisy conditions. Wang et al. [19] put forward a novel approach for rolling bearing fault diagnosis utilising BLS and CNN with attention mechanism, employing short-time Fourier transform to convert one-dimensional time-domain signals for model input, and attained favourable diagnostic outcomes in noisy conditions. The above research has achieved exceptional results in applying deep learning to fault identification; nonetheless, certain issues persist.

- (1) Transforming the raw vibration signal into a frequency domain signal can enhance the anti-noise and robustness of model, but some sensitive information may be lost during the conversion process, thereby reducing the accuracy of fault classification.
- (2) Convolutional neural networks can adaptively obtain signal features and have high identification accuracy. Nonetheless, the utilisation of CNN for feature extraction may result in the loss of certain feature information, hence diminishing the accuracy of problem identification.
- (3) When the collected vibration signals are noise free or the data distribution of training and testing samples is the same, some existing fault diagnosis methods can achieve significant classification results. However, rolling bearings often experience noise interference or exhibit covariate deviations between training and testing samples in real-world working environments, which makes it challenging to ensure the accuracy and stability of bearing fault classification models.

To solve the above problems, this paper proposes a rolling bearing fault diagnosis method based on DSICNN to improve the diagnostic performance of the model under strong noise and variable working conditions. The main contributions of this paper are as follows:

- (1) The original vibration signal and the vibration signal after Fast Fourier Transform are both used as inputs for the model, which preserves sensitive features more effectively and enhances the noise immunity of the model.
- (2) Dynamic weighted channel attention mechanism (DWCAM) and aggregated spatial self-attention mechanism (ASSAM) are put forward to adaptively capture the importance of different channels and regions of input information, thereby enabling CNN to extract fault features more fully.
- (3) The AT-LU activation function is constructed to improve the mandatory and rareness of the ReLU activation function when a negative input signal is involved. Moreover, the dual attention mechanisms and AT-LU are applied to the DSICNN model to achieve more outstanding generalisation and robust performance under strong noise and variable working conditions.

2 The proposed fault diagnosis method

2.1 Dynamic weighted channel attention mechanism

In convolutional neural networks, each channel of the feature map is activated by a corresponding convolutional filter response. Typically, each channel in the feature

map represents a feature, and the importance of different channels varies. Therefore, this paper proposes a dynamic weighted channel attention mechanism (DWCAM), as shown in Figure 1, which can dynamically calibrate the channel weights of different inputs.

First, the parallel input feature maps are fused in the channel dimension. Then, the fused feature map is squeezed and excited by two 1×1 convolutions, which can not only effectively obtain the importance of each feature and assign weight values to each channel but also reduce the large amount of computational redundancy caused by the use of fully connected layers. Next, the Sigmoid function is utilised to assign weights to the channel. Finally, the information is enhanced and the channel weights are obtained by Softmax. The expression is shown in Equation 1

$$(W_{c1}, W_{c2}) = SM(\delta(BN(PCnv(BN(PCnv(Concat(x_1, x_2))))))) \quad (1)$$

where x_1, x_2 are the input data, $PCnv(\cdot)$ is pointwise convolution operation, $BN(\cdot)$ represents batch normalisation, $\delta(\cdot)$ is Sigmoid activation function, $SM(\cdot)$ is Softmax function, W_{c1} and W_{c2} represent channel weights.

2.2 Aggregated spatial self-attention mechanism

In order to adaptively integrate local and global features of input information, enhance the perception ability of the model and improve the diagnostic accuracy and robustness of the model, this paper constructs an aggregated spatial self-attention mechanism (ASSAM). The attention mechanism encodes more extensive contextual information into local features and automatically captures important region features, thereby enhancing feature expression ability and assigning greater weight to important regions, as shown in Figure 2. After extracting features from the first branch, it is reshaped into $C \times WH$, and the expression is as follows.

$$S^A = RS(PCnv(x)) \quad (2)$$

where x is the input data, $RS(\cdot)$ represents reshaping data operation,

In the second branch, global average pooling is used to retain the background information of the input feature map in the channel direction, and then it is reshaped to $C \times 1$, as shown in formula (3).

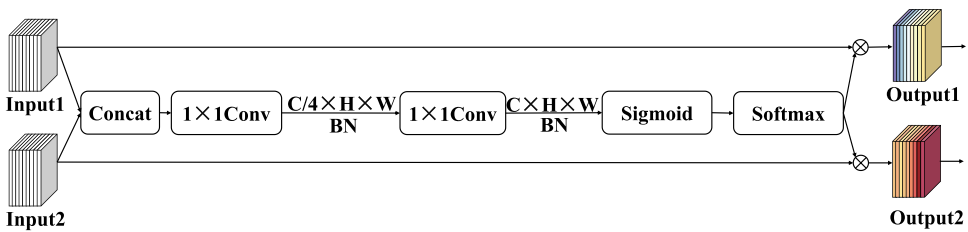


Figure 1. Dynamic weighted channel attention mechanism.

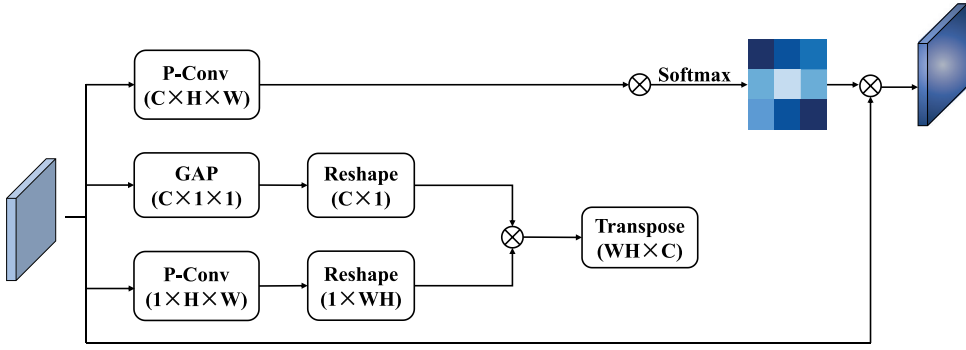


Figure 2. Aggregated spatial self-attention mechanism.

$$S^1 = RS\left(\frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W x(i, j)\right) \quad (3)$$

In the third branch, the pointwise convolution is exerted to enhance the feature information of the input feature map and maintain its intact in the W and H dimensions and reshape it to $1 \times WH$. The formula is described as follows:

$$S^2 = RS(PCov(x)) \quad (4)$$

Then the matrix multiplication is performed with the output of the second branch and transposed to multiply by the output of the first branch, the calculation is as follows:

$$S^B = TS(S^1 \times S^2) \quad (5)$$

where $TS(\cdot)$ is transpose operation.

The result of multiplication increases the dynamic range of attention through the Softmax function and assigns different weights to different regions of the input feature map to obtain a spatial attention map. The multiplication of the matrix is equivalent to establishing connections between each pixel, and calculating the similarity between two positional feature representations. The higher the similarity, the greater the weight assigned. Finally, the obtained spatial attention map is multiplied with the input feature map to complete the feature recalibration, and the calculation formula is as follows (6):

$$S = x \times \frac{\exp(S_i^A \times S_j^B)}{\sum_{i=1}^N \exp(S_i^A \times S_j^B)} \quad (6)$$

where S_i^A is the i th row of the feature map generated by the first branch, and S_j^B is the j th column of the feature maps generated by the second and third branches.

2.3 AT-LU activation function

The selection of activation function has a crucial influence on the training process of neural network. An appropriate activation function can accelerate the learning speed of the model and make the model converge better, so as to improve the classification performance of the model. The traditional activation function is prone to gradient

vanishing when applied to practical classification models. The proposal of ReLU activation function effectively solves the problem, as shown in Figure 3, and the function expression is shown in formula (7).

$$f(x) = \begin{cases} x, & x \geq 0 \\ 0 & x < 0 \end{cases} \quad (7)$$

From Figure 3, it can be seen that when $x > 0$ the ReLU activation function does not cause gradient saturation or vanishing problems and improves the convergence speed of the model. However, when $x < 0$, due to the constant gradient of the function being 0, neuronal necrosis occurs, which causes some neuron parameters not to be updated, and the effective information of this part of the data cannot be extracted. In order to address this issue, this paper proposes an AT-LU activation function, and its function expression is as follows:

$$f(x) = \begin{cases} x & x \geq 0 \\ \arctan(\frac{e^x-1}{e^x+1}) & x < 0 \end{cases} \quad (8)$$

Figure 4 is the graph of the AT-LU activation function. The activation function preserves the linear characteristics of the ReLU function at $x \geq 0$, which solves the problem of gradient disappearance during model training. And when $x = 0$, the function has both negative output values and soft saturation, which not only makes the overall output of the function close to 0 and improves the convergence speed of the model but also has stronger robustness in the presence of input changes or noise. The derivative of the activation function when $x < 0$ is shown in Equation 9:

$$f'(x) = \frac{e^x}{e^{2x} + 1} \quad (9)$$

When $x < 0$, $0 < e^{2x} < e^x < 1$, when $x = 0$, $f'(x) = \frac{1}{2}$. Thus, when $x < 0$:

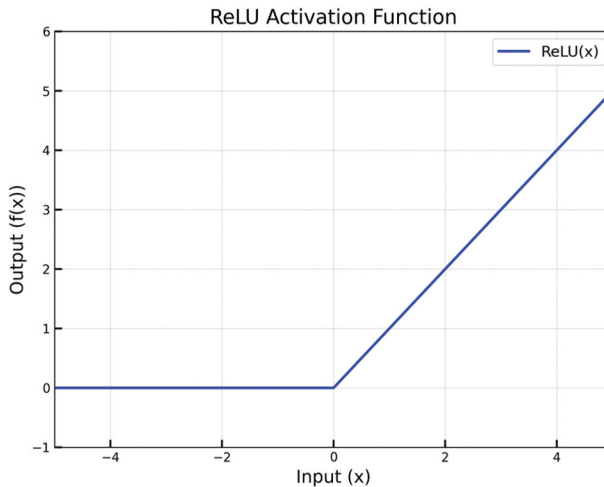


Figure 3. ReLU activation function image.

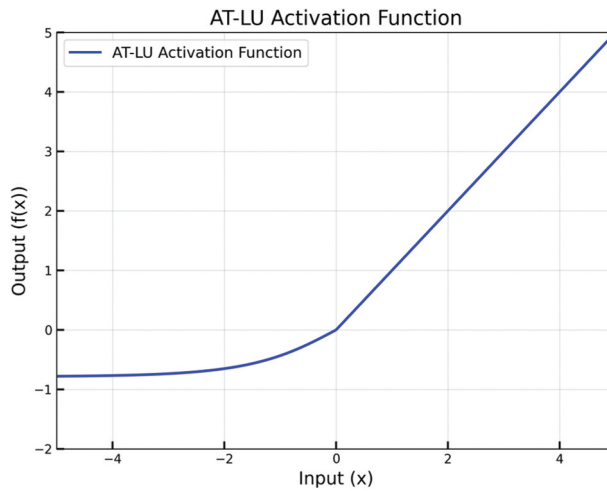


Figure 4. AT-LU activation function image.

$$0 < f'(x) < \frac{1}{2}$$

In summary, when the input of the neuron is negative, the gradient of the activation function still has continuous changes, which allows some network parameters of the negative input signal to be updated, thereby improving the learning ability of the model. At the same time, the participation of negative input signals alleviates the mandatory and scarcity of the ReLU activation function, which enables the model to learn the characteristics of the input signal more fully, and improves the learning performance and robustness of the model effectively.

2.4 The structure of DSICNN

In this paper, a dual-stream interactive convolutional neural network (DSICNN) is constructed, and the model structure is shown in Figure 5. In order to avoid the internal covariation offset and accelerate the convergence during model training, Batch Normalization (BN) [12,20,21] has been applied to the DSICNN model. The model has two inputs: the original signal and the spectral signal obtained through FFT. In order to fully extract fault features from the signal, the parallel input signal is subjected to two convolutional layers and BN operations simultaneously, and the extracted features are interacted with each other. The dilated convolution is used to extract the post-interactive features to expand the receptive field and capture richer contextual information. Afterwards, the max pooling layer is adopted to extract texture information, which can reduce model size, improve model training speed and alleviate the excessive sensitivity of convolutional layers to position while retaining useful information. After passing through the BN layer, the feature information is sent to DWCAM and ASSAM to highlight important information in the channel and space. Finally, after further extracting fault feature information by using dilated convolution layer, convolution layer and

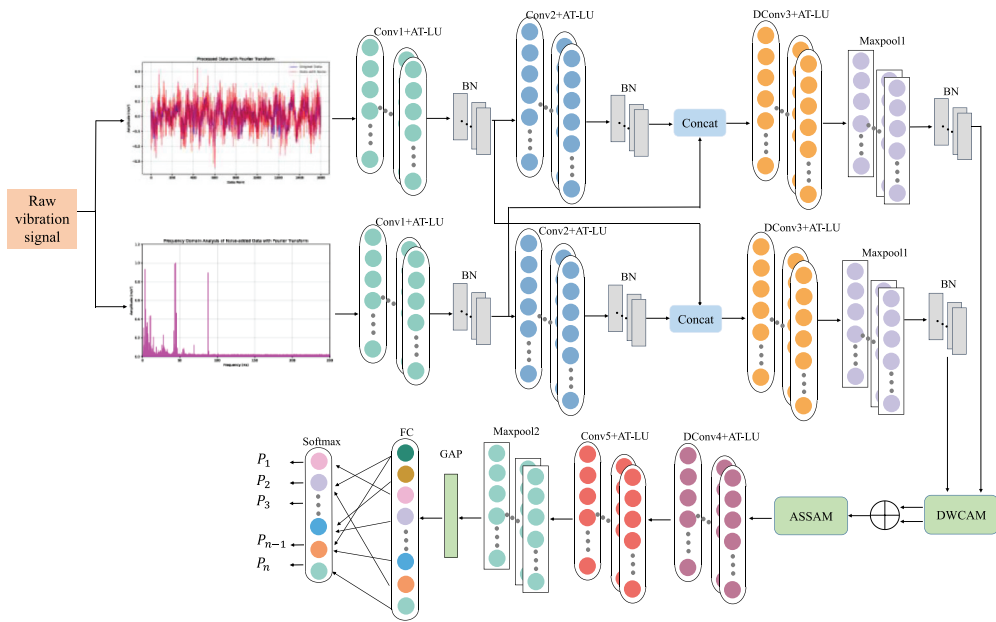


Figure 5. Structure of DSICNN.

maximum pooling layer, it is flattened and integrated for fault classification. The structural parameters of DSICNN are shown in Table 1, where the AT-LU activation function is selected for the convolutional layer.

3. Experimental validation and analysis

3.1 Experimental data sets

In this paper, the rolling bearing data set of Case Western Reserve University (CWRU) [22] and the MFS test data of our laboratory are used to verify the effectiveness of the fault diagnosis method proposed. As shown in Figure 6, the CWRU rolling bearing experimental platform consists of a motor, a drive end bearing, a torque sensor and a load. The bearing type is SKF6250 deep groove ball bearing, and the bearing defect is a single-point defect formed by electrical discharge machining. Under 1-3hp load, the

Table 1. Structural parameters of DSICNN.

Feature level	Structural parameters	Output size
Input	/	(1024, 1)
Conv1	(3, 16, 2)	(512, 16)
Conv2	(5, 16, 1)	(512, 16)
DConv1	(32,5,1,dilation rate=2)	(512, 32)
Maxpool1	(3,1)	(512, 32)
DConv2	(32,3,1,dilation rate=3)	(512, 32)
Conv3	(3,64,2)	(256, 64)
Maxpool12	(2,2)	(128, 64)
GAP	/	(64)
FC	Units=100, activation=ReLU	(100)
Softmax classifier	/	7

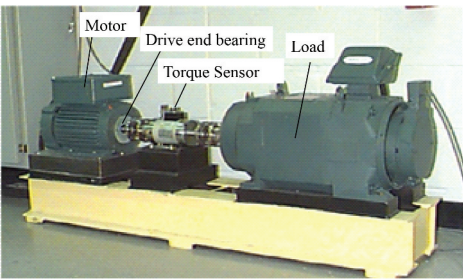


Figure 6. CWRU bearing test rig.

inner ring, outer ring and rolling element fault signals are collected at a sampling frequency of 12 kHz. In addition to the normal state, the diameters of the defects are 0.18 and 0.36 mm, respectively, with a total of seven fault categories.

The MFS bearing fault test platform of our laboratory is shown in Figure 7. The drive end is a faulty bearing and the remote end is a healthy bearing. The bearing type is ER-16K deep groove ball bearing. The fault of the bearing is processed using laser etching technology, with a fault depth of 0.25 mm. The fault diameter of the rolling element and the fault width of the inner and outer rings are 0.6 and 1.2 mm, respectively, as shown in Figure 8. The experimental platform applied a radial load of 50 N to the shaft using a 5.1 kg rotor disk mounted between two bearings. In the experiment, the industrial monitoring instrument (IMI) accelerometer 608A11 is used to record the vibration signals, and the shaft speed is monitored by the tachometer. At the same time, the multi-channel

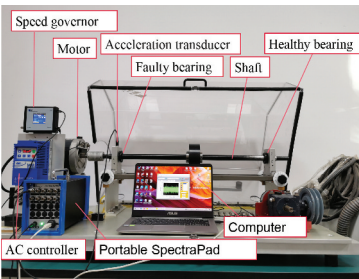


Figure 7. MFS bearing test rig.

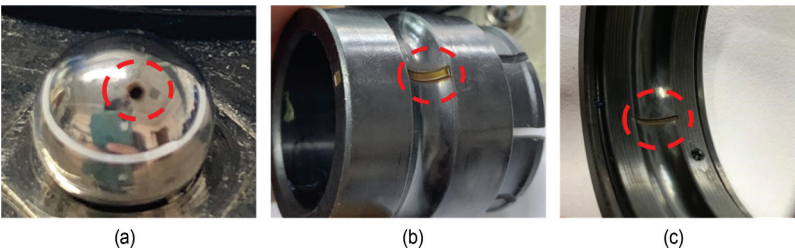


Figure 8. Failure location of ER-16K bearing. (a) Rolling element fault; (b) IF; (c) OF.

Table 2. Description of CWRU rolling bearing data set.

		0		0.18		0.36			
		None	Inner race	Outer race	Ball	Inner race	Outer race		
Fault diameter/mm		None	Inner race	Outer race	Ball	Inner race	Outer race	Ball	
Labels		0	1	2	3	4	5	6	Load
working condition A	training set	150	150	150	150	150	150	150	1hp
	test set	100	100	100	100	100	100	100	(0.7457kW)
working condition B	training set	150	150	150	150	150	150	150	2hp
	test set	100	100	100	100	100	100	100	(1.4914kW)
working condition C	training set	150	150	150	150	150	150	150	3hp
	test set	100	100	100	100	100	100	100	(2.2371kW)

Table 3. Description of MFS rolling bearing data set.

		0		0.6		1.2		Rotational speed/ r·min ⁻¹	
		None	Inner race	Outer race	Ball	Inner race	Outer race		Ball
Fault diameter/mm	Labels	0	1	2	3	4	5	6	
working condition D	training set	150	150	150	150	150	150	150	1200
	test set	100	100	100	100	100	100	100	
working condition E	training set	150	150	150	150	150	150	150	1300
	test set	100	100	100	100	100	100	100	
working condition F	training set	150	150	150	150	150	150	150	1400
	test set	100	100	100	100	100	100	100	

SpectraPad data acquisition device is used, and the signal processing and analysis is carried out in VQPro software. The sampling frequency is 15.36 kHz, and the motor speeds are 1200 r·min⁻¹, 1300 r·min⁻¹ and 1400 r·min⁻¹, respectively.

In this study, the sample length is set to 2048. One hundred and fifty and 100 samples are chosen as the training and test set per fault, respectively. The test datasets for each working condition of CUWR and MFS are shown in [Tables 2](#) and [3](#), respectively.

3.2 Fault diagnosis under different noise environments

In practical industrial applications, due to the noisy working environment of rolling bearings, the collected vibration signals often contain some noise, which makes it difficult to extract bearing fault information and thus affects the performance of model fault diagnosis. To verify the noise resistance of the proposed model, Gaussian noise with different signal-to-noise ratios (SNRs) is added to the original signal.

3.2.1 Fault diagnosis of CWRU dataset

In order to verify the effectiveness of the DSICNN model, in this experiment, the proposed method is compared with the deep learning models 1D-DCNN [14], 1D-CNN [17], TFRCNN [23] and MSD-CNN [24]. The diagnostic performance of the model under different noise environments and loads is tested using the CWRU dataset. To avoid experimental randomness, each group of experiments is repeated 10 times to calculate the average value. The experimental results are shown in [Table 3](#).

Table 4. Experimental results of each method under different loads and SNRs.

Algorithm		DSICNN	TFCNN	1D-DCNN	MSD	1D-CNN
Noise intensity		Accuracy				
0	A-A	99.87	98.29	97.46	99.39	99.33
	B-B	99.83	98.47	98.11	99.1	98.8
	C-C	99.64	98.2	98.97	99.07	99.09
-4	A-A	98.77	96.76	96.06	98.66	96.57
	B-B	98.89	97.3	96.27	98.44	98.16
	C-C	98.57	96.54	96.83	98.2	97.17
-6	A-A	97.1	87.17	94.26	93.43	95.54
	B-B	97.55	87.71	92.37	95.44	96.07
	C-C	97.36	86.95	93.9	94.3	94.2
-8	A-A	95.26	82.59	89.3	90.32	93.09
	B-B	95.04	82.36	89.99	89.8	93.76
	C-C	95.53	80.47	90.4	89.93	92.26
-10	A-A	93.97	77.84	87.44	84.17	89.29
	B-B	93.17	77	87.6	83.87	89.84
	C-C	92.88	78.6	85.37	84.8	90.64

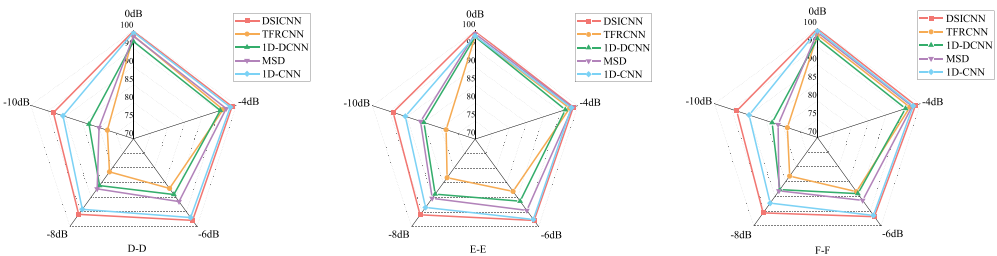


Figure 9. Radagrams for different SNRs under experiments D-D, E-E and F-F.

It can be seen from Table 4 that although the four comparison methods have good fault diagnosis performance when the SNR is high, as the SNR decreases, the diagnostic accuracy of the four comparison models fluctuates greatly, and the robustness performance is poor. However, the diagnostic accuracy of DSICNN is good in different noise environments. When the SNR is 0 dB, the classification accuracy of DSICNN reaches more than 99.64%, which is the highest among the five methods. When the SNR reduces to -10 dB, the fault diagnosis accuracy of the DSICNN model still reaches over 92.88%, which is 2.24%–16.97% higher than other methods. Taking the experiment A-A as an example, the classification accuracy of DSICNN is 99.87% when the SNR is 0, and it can still maintain a classification accuracy of 93.97% when the SNR is -10, which is significantly higher than other comparison methods. Therefore, DSICNN has stronger noise immunity and higher fault diagnosis accuracy.

3.2.2 Fault diagnosis of MFS dataset

In this experiment, the MFS bearing data set of our laboratory is used to test the fault diagnosis performance of the model proposed at different speeds and SNRs, and the results are visualised by radar map, as shown in Figure 9. From Figure 9, it can be seen that the DSICNN model not only has higher fault classification accuracy under different SNRs but also has a smaller decrease in diagnostic accuracy with the increase

in noise. Taking the data set E-E as an example, when the SNR is -10 dB, the fault classification accuracy of the DSICNN model is 93.95%, which is 3.59%–15.49% higher than the other four classification methods. With the decrease of noise, the diagnostic accuracy of all methods gradually improves but the stability of DSICNN model is more excellent. When the SNR increases from -10 dB to 0 dB, the classification accuracy of the DSICNN model increases by 5.95%, and the classification accuracy of other methods enhances by 8.18%–19.97%. Therefore, it can be demonstrated that the DSICNN model has marvellous robustness and accuracy under strong noise.

3.3 Diagnostic results under variable operating conditions and strong noise environment

In practical engineering, the working environment of the bearing often has noise interference and complex and variable working conditions, which affect the fault recognition accuracy of the model. Therefore, this experiment tests the diagnostic performance and generalisation ability of the model under the noise of -4 dB and variable load conditions.

3.3.1 Experiments on bearing data sets under variable load conditions

In order to prove the effectiveness of the DSICNN model under variable load, this section tests its fault recognition accuracy under the CWRU data set. The experimental result is shown in Figure 10. In Figure 10(a,b) indicates that the A and B data sets are training samples and test samples, respectively. The average recognition accuracy of the model in this paper is 98.48% under six different working conditions, and higher than that of other comparative models. Among the six different operating conditions, the fault recognition accuracy of C-B is the highest, at 98.69%, which is 2.23%–11.95% higher than other

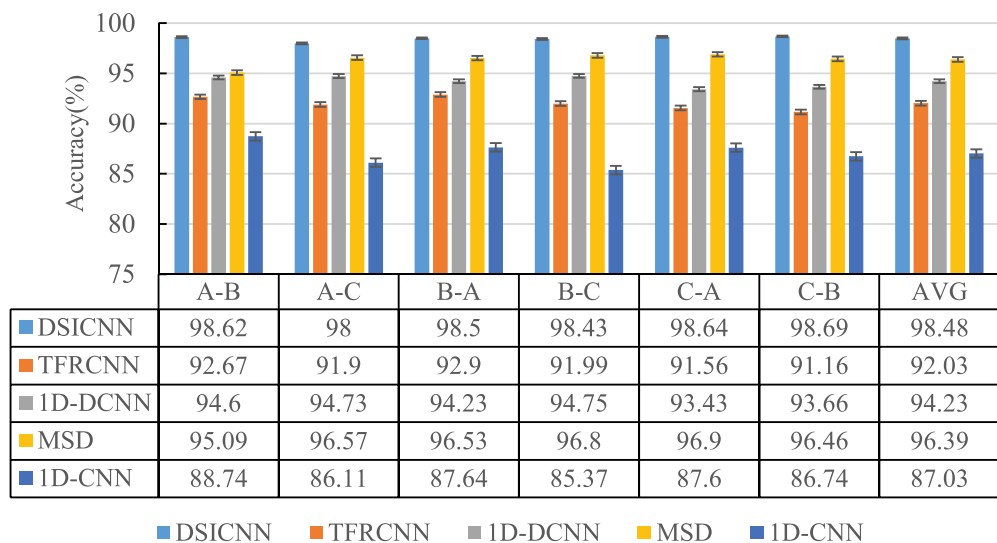


Figure 10. Fault diagnosis of variable load under strong noise environment.

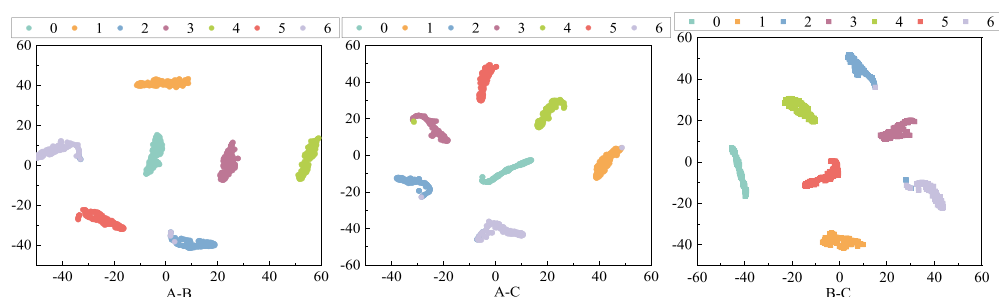


Figure 11. T-SNE visualisation.

comparative models. The lowest accuracy is 98% at A-C condition, which is 1.43% higher than the highest in the comparison model. In addition, from the error lines in the figure, it can be seen that DSICNN has the best stability with smaller error and stronger anti-noise performance. In summary, DSICNN has better fault diagnosis performance, stronger generalisation ability and stability in noisy environment.

By using the t-SNE nonlinear dimensionality reduction algorithm, the feature extraction and fault classification capabilities of the DSICNN model at -4 dB can be more intuitively demonstrated. The classification results of A-B, A-C and B-C are shown in Figure 11. It can be seen from the graph that the samples classified by the DSICNN model are mostly clustered, with only individual samples being misclassified. In addition, there is a significant distance between the classification results of different categories, and the samples are scattered, which can better distinguish various fault types. Therefore, the DSICNN model still has strong fault classification ability under strong noise and variable load.

3.3.2 Experiments on bearing data sets under variable speed conditions

In this section, the classification performance of the DSICNN model under variable speed conditions is tested using the MFS bearing data set of our laboratory. The experimental results are shown in Figure 12.

From Figure 12, it can be observed that the recognition accuracy of DSICNN is tremendously higher than that of the other four methods at six different speeds. The average recognition accuracy is 98.46%, which is 2.34% higher than the MSD with the highest recognition accuracy in the comparison method, and the variance is smaller, indicating that the method proposed has stronger anti-interference ability in noise environment. Under D-E condition, the recognition accuracy of DSICNN is 98.57%, which is 1.84%–9.15% higher than other comparison methods. Meanwhile, it can be seen from the error line in Figure 12 that the model proposed has better robustness and more stable fault recognition performance. In summary, the model proposed also exhibits good fault classification accuracy and noise immunity under variable speed condition.

With a view to more intuitively seeing the number and types of bearing samples misclassified, a confusion matrix is introduced to visualise the classification results. Figure 13 shows the fault classification results under D-E, F-D and F-E conditions. Taking Figure 13(a) as an example, in -4 dB noise environment, only 10 out of the 700 samples are misclassified, with the 0.18 mm inner ring fault being misclassified as normal

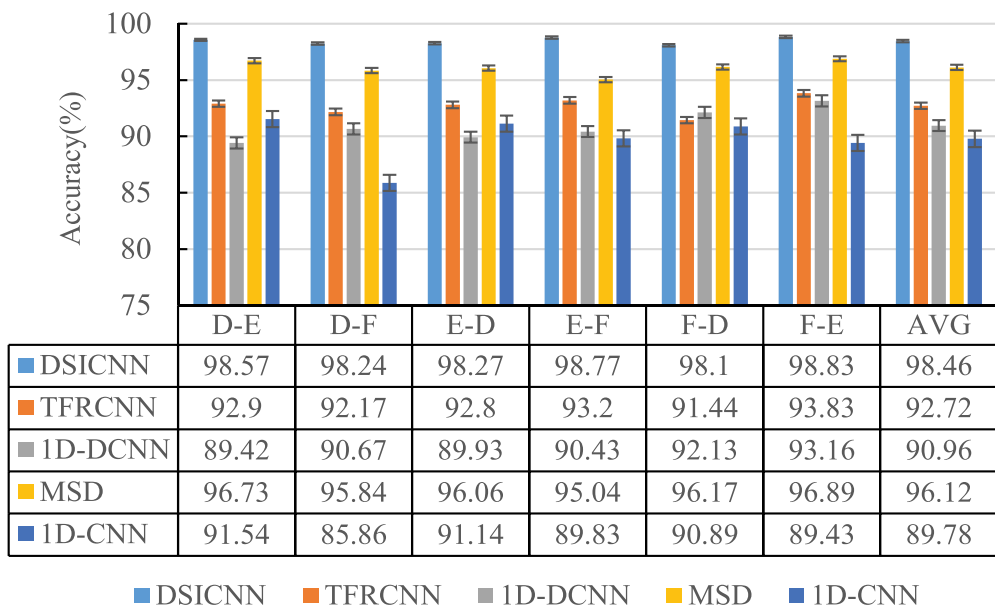


Figure 12. Fault diagnosis of variable speed under strong noise environment.

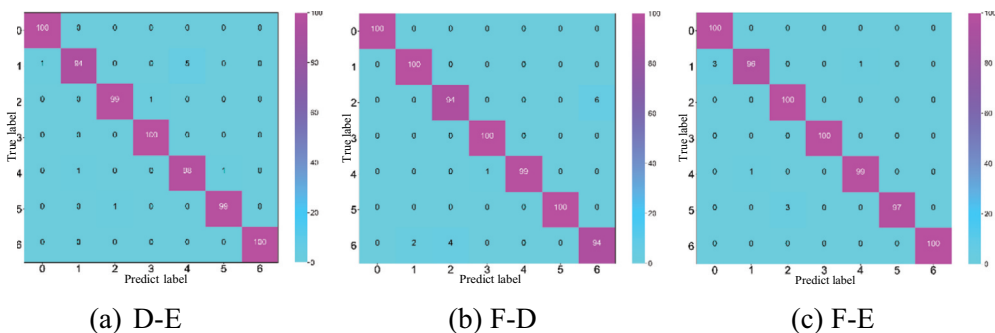


Figure 13. Confusion matrix.

and the 0.36 mm inner ring fault. At the same time, DSICNN also has superior classification performance under other working conditions. Therefore, the model proposed in this paper also has good fault diagnosis performance and generalisation ability under variable speed.

3.4 Ablation experiment

For the sake of verifying the superiority of the model proposed, ablation experiment is conducted under -4 dB and six different working conditions. The comparison models include CNN (convolutional neural network model with only original signal as input), DSICNN1 (Adding FFT transformed input information to CNN), DSICNN2 (Adding

Table 5. Ablation experimental results.

Algorithm	CNN	DSICNN1	DSICNN2	DSICNN3	DSICNN
work conditions	Accuracy (%)				
A-A	85.66	89.10	93.47	96.53	98.77
B-B	84.38	91.18	92.87	96.44	98.89
C-C	86.62	90.54	93.59	97.01	98.57
D-D	84.63	92.00	94.29	96.30	98.77
E-E	85.91	91.95	94.06	96.13	98.72
F-F	84.60	91.34	95.88	96.47	98.62
Average	85.30 $\pm$ 0.90	91.02 $\pm$ 1.08	94.03 $\pm$ 1.03	96.48 $\pm$ 0.30	98.72 $\pm$ 0.12

interactive operations to DSICNN1), DSICNN3 (Adding DWCAM to DSICNN2) and DSICNN. The experimental results are shown in Table 5.

It can be observed from Table 5 that the classification accuracy of the CNN model is the lowest under all working conditions. DSICNN1 retains the sensitive information in the time domain signal while acquiring the frequency domain feature information. The interactive operation enables DSICNN2 to have stronger processing ability for the noise signal. Therefore, with the gradual addition of the frequency domain signal and the interactive operation, the diagnostic accuracy of DSICNN1 and DSICNN2 in the noise environment is significantly improved. Moreover, DSICNN3 and DSICNN have improved fault classification accuracy while significantly enhancing robustness. This is because DWCAM dynamically calibrates the channel weights of different inputs, highlighting the important fault features of the channel dimension, and ASSAM integrates broader context information into local features and automatically captures important regional features. By comparing the five models, it can be concluded that DSICNN has the most outstanding anti-noise performance and stability, because the two-stream network model, interactive operation and dual attention mechanisms all have positive effects on the model.

4. Conclusions

In this paper, a rolling bearing fault diagnosis method based on DSICNN is proposed to improve the classification performance of traditional fault diagnosis models under strong noise and variable working conditions. The method is validated by CWRU and MFS rolling bearing data sets in our laboratory. The specific conclusions are as follows:

- (1) The time domain signal and the frequency domain signal transformed by FFT are used as inputs of the neural network, which not only fully extracts sensitive feature information from the original signal but also weakens the interference of noise and improves the anti-noise performance of the model.
- (2) A dynamic weighted channel attention mechanism (DWCAM) and an aggregate spatial self-attention mechanism (ASSAM) are constructed to enable the model to automatically capture the important information of the feature map's channel and position. At the same time, the AT-LU activation function has negative output value and soft saturation, which alleviates the mandatory scarcity of the ReLU activation function and effectively improves the learning performance and robustness of the model.

- (3) A dual-stream interactive convolutional neural network (DSICNN) is put forward to interact with the extracted time-domain and frequency-domain features, extract fault features more comprehensively and improve the generalisation ability and robustness of the model.
- (4) The experimental results using two datasets show that the method rendered in this paper has outstanding diagnostic performance, generalisation performance and robustness in the case of strong noise and variable working conditions.

In industrial applications, the interpretability of fault diagnosis systems is crucial, especially in ensuring system decision transparency. Therefore, in future research, methods such as SHAP (Shapley Additive exPlanations) or Grad CAM will be used to visualise the importance of features, to further improve the interpretability of the fault model.

Disclosure statement

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Data availability statement

The data cannot be made publicly available upon publication because they contain sensitive personal information. The data that support the findings of this study are available upon reasonable request from the authors.

Ethical considerations

This article does not include any studies with human or animal participants.

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